

SE LAB 1

Problem Statement 60 - Podcast Guest Scheduling and Outline Builder

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1. Requirements Table

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Guest can pick an open interview slot the host has put up, and submit talking points as bullet points plus a link to their bio.	High	Pass - slot gets booked, invite sent, outline shows up attached to the episode. Fail - guest manages to book a date the host had blocked.	This is the main feature basically, guests need a way to book without emailing back and forth a bunch of times.
FR-002	Functional	Host can set which dates/times are open for booking, and block dates they dont want guests booking on.	High	Pass - new slots show up for guests right away, blocked dates cant be picked. Fail - a blocked date is still bookable somehow.	Host needs to be in control of their own calendar or the tool is kind of pointless for them.
FR-003	Functional	When someone books or cancels, system should send a calendar invite and confirmation email to both guest and host on its own.	Medium	Pass - both people get invite/email within like a minute of booking. Fail - one of them doesnt get it.	If its manual people will forget, so this has to be automatic.
FR-004	Functional	Guest should be able to reschedule or cancel, but only up to 24 hrs before the interview, and the slot opens back up after.	Medium	Pass - slot becomes available again right after a valid cancel/reschedule. Fail - slot stays blocked or host doesnt find out about the change.	Gives guest some flexibility but still protects the host from last minute cancellations.
FR-005	Functional	Host can generate a PDF sheet that puts together all the guest outlines for an episode with timestamps, ready before recording starts.	High	Pass - PDF has all outlines in the right order with timestamps. Fail - an outline is missing or timestamps are messed up.	Saves the host from copy pasting notes from different guests manually every time before recording.

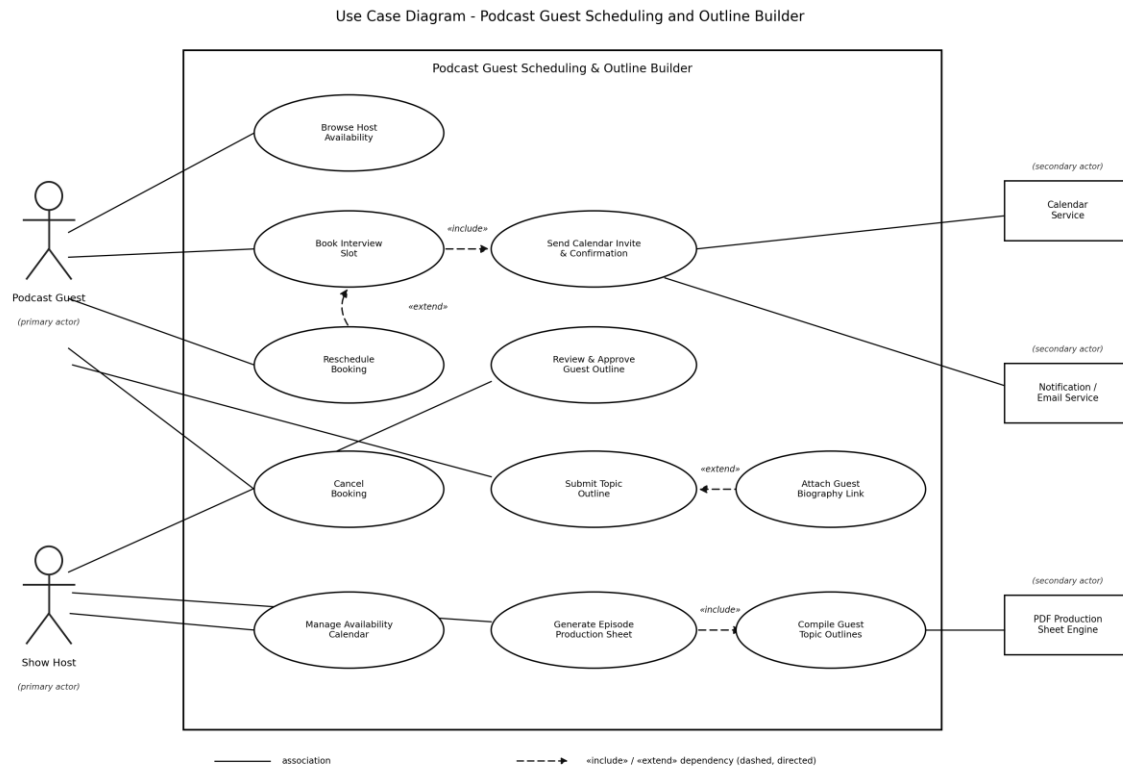
Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance and Security	PDF generation should take under 1 second, and guest info (bio link, email etc) should be encrypted, not just stored as plain text somewhere.	High	Pass - PDF generates in under 1 sec even with like 10 guests, data is encrypted. Fail - takes longer, or data isnt encrypted.	Hosts do this right before recording so it needs to be quick, and guest info is personal so it should be safe.
NFR-002	Usability and Reliability	Booking page should work fine on mobile, load in under 2 seconds, and be reasonably accessible. System should basically always be up.	Medium	Pass - page loads under 2s on mobile, passes a basic accessibility check, uptime stays above 99.5%. Fail - any of these not met.	Most guests will probably book from their phone while doing something else

ID	Type	Description	Priority	Acceptance Criteria	Rationale
					so it needs to be fast and simple.

2. UML Use Case Diagram

Primary actors are Podcast Guest and Show Host. Secondary/system actors are Calendar Service, Notification/Email Service and the PDF Production Sheet Engine. Include used - Book Interview Slot includes Send Calendar Invite and Confirmation, and Generate Episode Production Sheet includes Compile Guest Topic Outlines. Extend used - Reschedule Booking extends Book Interview Slot, and Attach Guest Biography Link extends Submit Topic Outline.



3. Use Case Flow Specification

Use Case: Book Interview Slot

Primary Actor - Podcast Guest

Secondary Actors - Calendar Service, Notification/Email Service (through the included use case), Show Host (gets notified once booking is done)

Preconditions

- Guest has an account and is logged in.
- Host has put up at least one open slot.
- The slot the guest wants isn't already taken.

Postconditions

- Success - slot is marked as booked, invite and confirmation sent to guest and host, and an empty outline gets created for the guest to fill in later.

- Failure - nothing is booked, guest just sees the updated list of open slots.

Main Success Scenario

1. Guest logs into the portal.
2. Guest goes to the hosts availability page.
3. System shows the open slots.
4. Guest picks a slot.
5. System checks the slot is still free.
6. Guest confirms the booking (name, email, topic).
7. System reserves the slot.
8. System runs the included use case Send Calendar Invite and Confirmation, which sends the invite and email to guest and host.
9. System shows a confirmation and asks guest to submit their topic outline next.
10. Use case ends.

Alternate Flow - Slot Gets Taken Before Confirming

This can happen at step 5, if someone else books the same slot right before the guest confirms.

- 5a. System finds out the slot isn't free anymore.
- 5b. System tells the guest the slot just got taken.
- 5c. System refreshes the list of open slots.
- 5d. Guest picks a different slot and flow goes back to step 4.

Postcondition for this flow - nothing gets booked under the old slot, guest just sees an updated list and can try again.